

Ans 8
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Nmap Live Host Discovery.

ARP, ICMP, TCP/UDP Ping scans:

Aim: To discover live hosts on a network using Nmap host discovery, by ARP, ICMP, TCP/UDP Ping scans.

ARP Host Discovery:

```
$ sudo nmap -PR -sn 10.200.6.0/24
```

O/P: starting Nmap 7.80 (<https://nmap.org>)

Nmap scan report for 10.200.6.1
Host is up
Nmap done, 256 IP addresses (1 host up)
Scanned in 10.51 seconds.

ICMP Host Discovery:

ICMP Echo (-PE):

```
$ sudo nmap -PE -sn 10.200.6.0/24
```

Nmap scan report for 10.200.6.1

Host is up (0.069s latency)

```
Nmap scan report for 10.200.6.50
```

Host is up (0.070s latency)

ICMP Timestamp (-PP)

```
$ nmap -PP -sn 10.200.6.0/24
```

Nmap scan report for 10.200.6.50

Host is up (0.069s latency)

Nmap scan report for 10.200.6.

Host is up (0.069s latency)

Nmap done: 250 IP addresses (2 hosts up)

Scanned in 8.44 seconds.

ICMP Address Mask (-PM)

\$ nmap -PM -sn 10.200.6.0/24
Nmap done: 256 IP addresses (0 hosts up)
scanned in 52.14 seconds.

TCP+UDP Ping Discovery:

i) TCP SYN Ping:

\$ nmap -PS -sn 10.200.6.0/24
Nmap scan report for 10.200.6.50
Host is up (0.070s latency)
Nmap done: 256 IP addresses (2 hosts up)
scanned in 33.58 seconds.

ii) TCP ACK Ping (-PA)

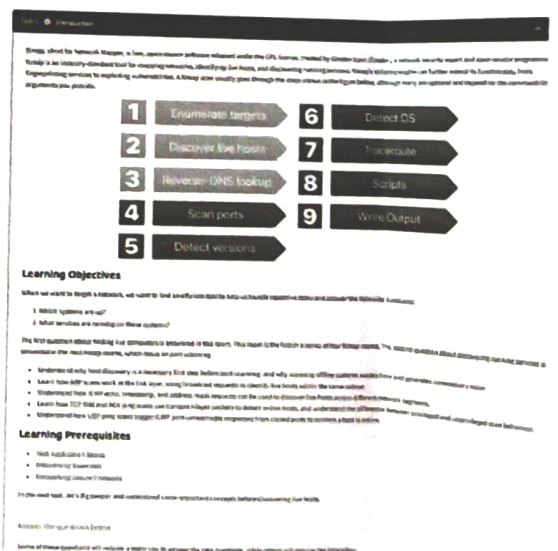
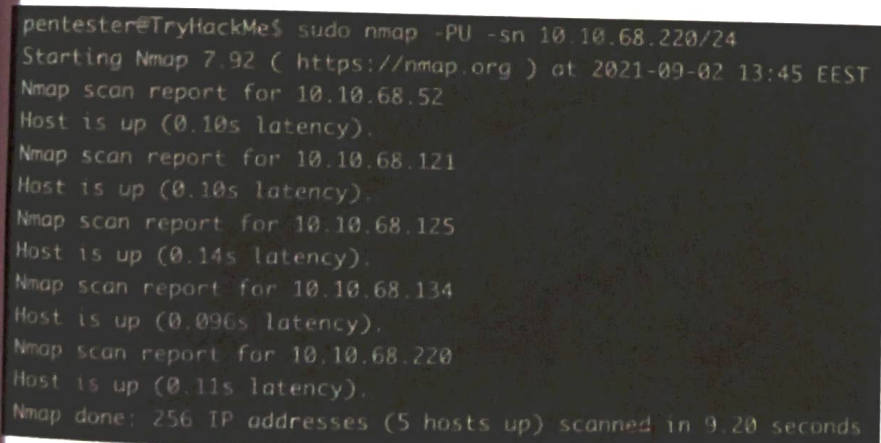
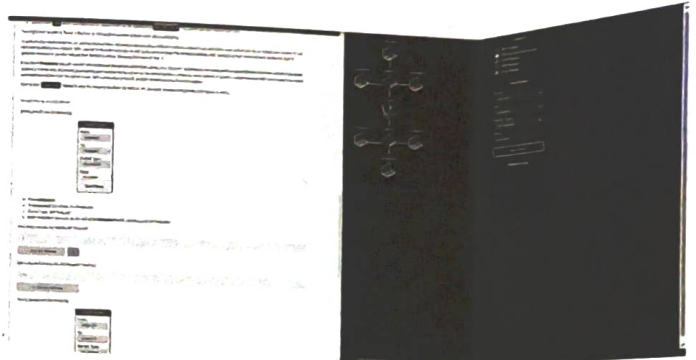
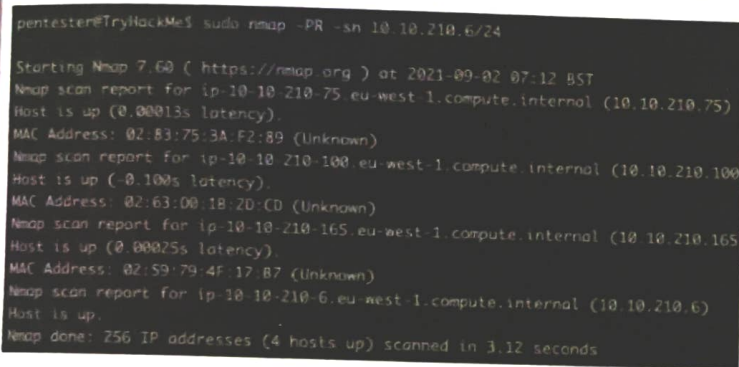
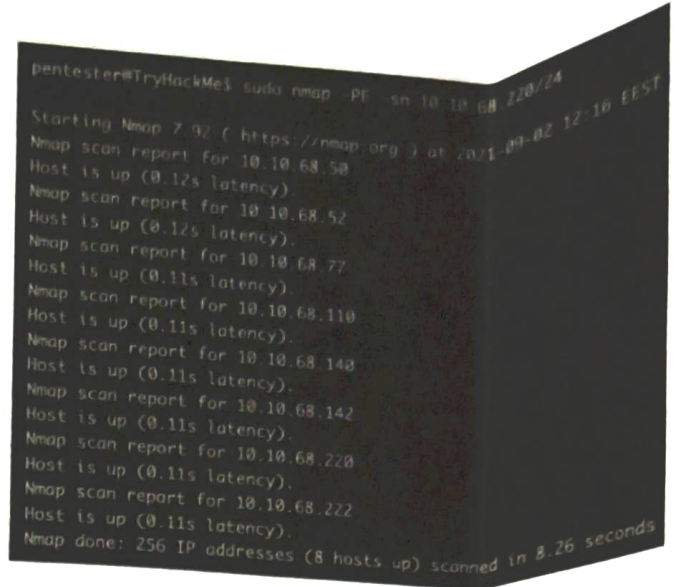
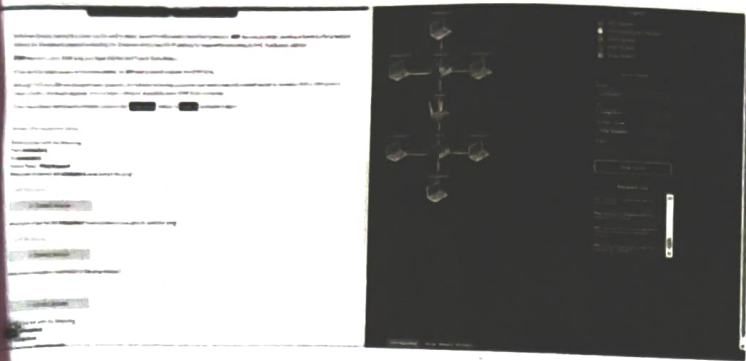
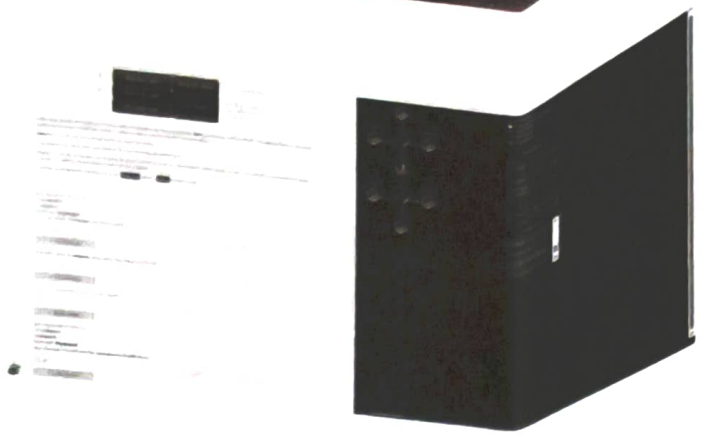
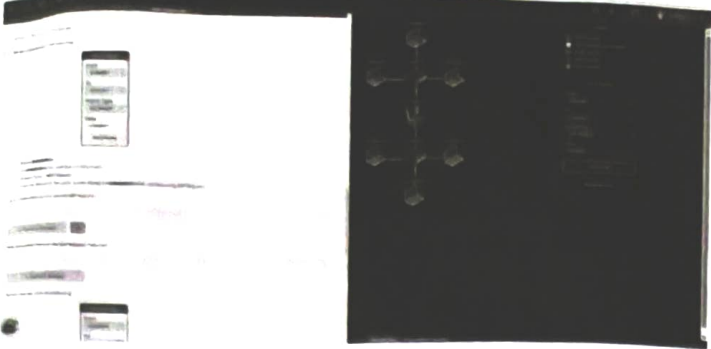
\$ sudo nmap -PA -sn 10.200.6.0/24
Nmap scan report for 10.200.6.50
Host is up (0.069s latency)
Nmap done: 256 IP addresses (2 hosts up)
scanned in 33.64 seconds.

iii) UDP Ping (-PU)

\$ sudo nmap -PU -sn 10.200.6.0/24
Nmap scan report for 10.200.6.50
Host is up (0.070s latency)
Nmap done: 256 IP addresses (2 hosts up)
scanned in 35.97 seconds.

Masscan - A Faster Alternative:

\$ masscan 10.200.6.0/24 -P 443
\$ masscan 10.200.6.0/24 -P 500,443
\$ masscan 10.200.6.0/24 -P 22-25



Result:

Thus line host discovery for
ARP ICMP, TCP, UDP was studied
and output got verified successfully